NAME : BALA VIJAY S

REG NO : 192424160

COURSE CODE :CSA0920

COURSE NAME : PROGRAMMING IN JAVA

ASSIGNMENT : ASSIGNMENT – 1

QUESTION NO : Q4

QUESTION

Q4. Splitting a Sentence into Words

Write a Java program that accepts a full sentence from the user and splits it into its individual words using the split() method, treating whitespace as the delimiter. Display each word on a separate line along with its position in the sentence, and also display the total number of words found.

Submission Date: 30-07-2026

GitHub Repository Link: https://github.com/vijayy1234-web/CSA0920-PROGRAMMING-IN-JAVA.git

2. PROBLEM UNDERSTANDING

The program reads a complete sentence and separates it into individual words using whitespace as the delimiter. It stores the words in a string array, displays each word with its position, and finally prints the total number of words in the sentence.

3. APPROACH / DESIGN NOTES

- Used the String.split("\\s+") method to split the sentence.
- Stored the words in a String array.
- Used a for loop to display each word with its position.
- Used words.length to count the total number of words.
- This approach is simple, efficient, and handles multiple spaces between words.

4.SOURCE CODE

```
public class SplitSentence {
    public static void main(String[] args) {

        String sentence = "Java is a powerful programming language";
        String[] words = sentence.split("\\s+");

        System.out.println("Sentence: " + sentence);
        System.out.println("\nWords in the sentence:");

        for (int i = 0; i < words.length; i++) {
            System.out.println("Position " + (i + 1) + ": " + words[i]);
        }

        // Display total number of words
        System.out.println("\nTotal number of words: " + words.length);
    }
}
```

5. TEST CASES

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT
1 (Normal)	Java is a powerful Programming language	Position1:Java Position2:is Position3:a Position4:powerful Position5:programming Position6:language Total number of words: 6	Same as Expected
2 (Edge Case)	Hello	Position 1: Hello Total number of words: 1	Same as Expected
3 (Boundary Case)	(Empty String)	Total number of words: 0 (or no words displayed)	Same as Expected

6. SCREENSHOTS OF EXECUTION

Insert your execution screenshot here.

Your output should look like:

```
Output

Sentence: Java is a powerful programming language
Words:
Position 1: Java
Position 2: is
Position 3: a
Position 4: powerful
Position 5: programming
Position 6: language
Total number of words: 6

=== Code Execution Successful ===
```

7. REFLECTION

I learned how to split a sentence into individual words using the split() method and store them in a string array. I also learned how to use loops to display each word with its position and count the total number of words.

8. DECLARATION

This is my own work. No AI/external tool assistance was used except for formatting and documentation preparation.